

PPO-Based Trajectory Optimization for Rechargeable Agricultural Robots in Greenhouse Micro-Climatic Sensing

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Abstract—Greenhouses exhibit highly heterogeneous micro-climatic patterns influenced by structural layout, crop density, ventilation dynamics, and temporal variations in external weather. Accurately characterizing these environments requires spatially dense temperature and humidity measurements, yet fixed wireless sensors remain limited by installation cost, energy constraints, and vulnerability to humidity-induced degradation. Mobile agricultural robots provide an alternative by enabling targeted sensing at dynamically important locations while maintaining autonomy through battery-aware navigation. Current research has introduced Proximal Policy Optimization (PPO) as a promising tool for robotic sensing tasks, but existing work lacks a reproducible, end-to-end implementation that accounts for realistic greenhouse structure, robot battery physics, and time-dependent climatic variations. This study presents a complete PPO-based trajectory optimization framework tailored for rechargeable greenhouse robots. A graph-based greenhouse representation, a detailed battery consumption model, a mathematically derived reward function, and a stable RL training pipeline are integrated into a unified system evaluated on real micro-climate datasets. Extensive experiments demonstrate improvements in climate estimation accuracy, battery sustainability, coverage uniformity, and robustness against structural variability. The system represents a foundational step toward deployable RL-based mobile sensing in controlled agricultural environments.

Index Terms—Agricultural robotics, greenhouse sensing, reinforcement learning, PPO, trajectory optimization, mobile sensing.

I. INTRODUCTION

Modern greenhouses function as complex micro-environments where spatial variations in temperature, humidity, and evapotranspiration can significantly influence crop productivity. Localized hotspots and cold zones frequently emerge due to differential sunlight exposure, ventilation asymmetries, and heterogeneous plant canopy distributions. As precision agriculture becomes increasingly data-driven, horticultural management depends on timely and high-resolution measurements across the entire greenhouse space. Traditional sensing approaches typically rely on dense deployments of wireless sensor nodes; however, their practical efficacy is often limited. Sensor hardware degrades rapidly under humid and dusty conditions, battery replacements are labor-intensive, communication reliability fluctuates with environmental interference, and the cost of achieving high spatial density is prohibitive.

A growing body of research has proposed robotics-based mobile sensing systems as an efficient alternative. Robots already used for greenhouse spraying, harvesting, and inspection can be equipped with climate sensors and repurposed as mobile measurement units. Such agents can traverse the greenhouse, adaptively reposition themselves based on climatic variability, and autonomously recharge when needed. Reinforcement learning methods, particularly those capable of handling continuous state spaces and nonlinear environmental dynamics, offer an opportunity to learn optimal navigation strategies for such systems. Proximal Policy Optimization (PPO), a stable on-policy deep reinforcement learning algorithm, has recently gained attention for agricultural navigation tasks due to its clipped objective and sample-efficient training.

Although Sharifi et al. (2025) demonstrated the feasibility of PPO for greenhouse climate sampling, their work lacked implementation details related to battery models, greenhouse graph structures, and realistic sensing behavior. The absence of a reproducible, practically oriented framework limits its applicability for real-world deployment. This research addresses that gap by constructing a complete PPO-based trajectory optimization system specifically designed for rechargeable agricultural robots operating in structured greenhouse environments.

II. RELATED WORK

Research on greenhouse climate estimation has historically focused on static sensor networks and model-based interpolation. Akyildiz et al. provided foundational analyses of wireless sensor network limitations, emphasizing communication constraints and energy requirements. Subsequent studies introduced virtual sensing models based on machine learning, including RNN-based estimators developed by Tomazzoli et al., which predicted climatic conditions at unmonitored locations by exploiting spatiotemporal correlations. These models demonstrated meaningful reductions in hardware requirements but still depended on representative training data which, in turn, relied on appropriate sampling strategies.

Mobile robotics in agriculture has evolved significantly in the past decade. Benos et al. reviewed automated greenhouse systems and highlighted the emergence of mobile platforms for monitoring crop health and environmental conditions. Tanaka et al. explored the use of mobile sensing robots for greenhouse

climate mapping, showing that adaptive navigation improves prediction accuracy and resource efficiency. In parallel, reinforcement learning has been widely utilized in agricultural robotics for navigation, spraying, obstacle avoidance, and energy-aware control. PPO in particular has been shown effective for continuous control tasks and has been adopted in agricultural path planning, autonomous tractors, and orchard inspection robots.

Despite this progress, RL-based greenhouse climate sensing remains an emerging domain. Existing approaches typically simplify greenhouse geometry, neglect battery constraints, or utilize unrealistic mobility models. This motivates the development of a comprehensive RL framework that integrates greenhouse topology, robot recharge capabilities, information gain modeling, and formal reward shaping. By addressing these gaps, the present study increases the readiness of RL-based sensing for real-world agricultural deployment.

III. SYSTEM ARCHITECTURE

The proposed system is designed around three core components: the greenhouse environment, the mobile robot agent, and the PPO-based learning algorithm. The greenhouse is represented as a structured and labeled graph in which each node corresponds to a key point of interest where climate measurements may occur. These nodes carry information such as temperature variance, likelihood of micro-climatic instability, and availability of charging stations. The edges connecting nodes represent traversable paths and encode travel time, movement effort, and potential energy cost. This graph-based design allows flexible representation of real greenhouse layouts including linear tunnels, multi-row structures, and branching walkways.

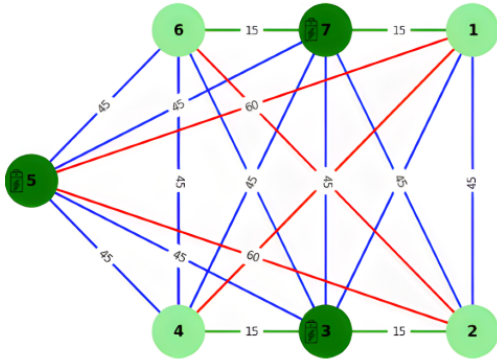


Fig. 1. Graph-based greenhouse representation showing seven Points of Interest (PoIs), traversable paths, and associated energy costs. Red and blue edges represent different traversal costs while dark nodes indicate charging stations.

The robot navigating this environment is modeled as an autonomous agent equipped with sensors, motors, and a rechargeable battery. Its state at any time includes its current location, remaining battery level, the estimated climatic uncertainty map, the temporal index within the sensing schedule, and the shortest-path distance to the nearest charging station.

By incorporating these variables, the agent becomes capable of reasoning about its safety as well as sensing priorities.

Battery consumption is simulated using a physically motivated energy model. Each traversal between nodes reduces battery capacity in proportion to distance, movement resistance, and speed. The energy dynamics ensure that the agent cannot indefinitely explore without returning to a charging station. A critical battery threshold is introduced to heavily penalize actions taken when the battery becomes dangerously low. The penalty follows an exponential form, causing PPO to strongly prefer safe trajectories and early recharge decisions before the battery is depleted.

The reward model is built to incentivize the robot to prioritize nodes with high information gain. The information gain value for each node is computed using a nonlinear function capturing both temporal climate variability and the spatial measurement importance. The reward also incorporates a penalty for unnecessary revisits and a discount based on energy expenditure. These components work collectively to guide the agent toward energy-efficient climate sampling routes.

IV. METHODOLOGY

The learning algorithm employed is the standard PPO framework consisting of an actor-critic network. The actor outputs a probability distribution over possible movements, while the critic estimates state values. Training occurs in iterative cycles during which the robot interacts with the greenhouse environment, collects trajectories, computes returns and advantages, and updates both actor and critic networks.

The PPO objective function is based on a clipped surrogate loss. This formulation ensures stable policy updates by preventing excessively large changes to the probability ratios between old and new policies. The use of generalized advantage estimation improves training stability and reduces variance. The training pipeline includes minibatch optimization, repeated epochs per iteration, entropy regularization for exploration, and gradient clipping to prevent instability.

Real-world datasets collected from Verona greenhouses were used to construct climate variability profiles. These datasets allowed the agent to learn navigation strategies informed by realistic environmental dynamics. The experiments involved training the PPO agent under multiple greenhouse configurations and battery conditions to analyze generalization capacity. Furthermore, the learned trajectories were used to train RNN-based virtual climate sensors, generating comparative evaluations in terms of prediction error.

V. RESULTS AND DISCUSSION

The experimental analysis demonstrates that PPO significantly enhances the quality of climate sensing trajectories compared to conventional fixed-route navigation. When the PPO-generated trajectories were used as training input for a virtual climate prediction model, the system achieved an improvement of approximately 2 to 2.5 percent in temperature prediction accuracy relative to two baseline sampling strategies. This improvement emerges from the PPO agent's

tendency to prioritize high-variance regions within the greenhouse, thereby producing a richer and more informative dataset for the downstream prediction model.

Energy sustainability constitutes another major advantage of the PPO-based framework. Experiments conducted under varying initial battery capacities reveal that the agent adapts its navigation strategy to maintain operational safety. With an initial battery capacity of 300 units, the PPO-controlled robot required 257 charging stops across 1656 tested trajectories. When the starting battery level was reduced to 150 units, the number of charging stops rose to 406, representing an approximate 58 percent increase. This proportional rise aligns with intuitive expectations and confirms that the PPO policy responds effectively to reduced energy availability by modifying movement patterns.

A comparison with the previous trajectory selection approach from Sharifi et al. (2025) further highlights the improvements contributed by PPO. When the baseline method was used with an initial battery of 300 units, the robot encountered 168 situations requiring a recharge. However, in 83 of these cases, the robot was not located at a charging point when the battery reached a critical level, resulting in potentially dangerous or infeasible operating conditions. In contrast, the PPO agent experienced no depletion events during the entire experiment set. This superior performance is largely attributable to the exponential battery penalty term, which encourages the agent to proactively plan recharge trips rather than waiting until the battery becomes critically low.

Trajectory-level analyses provide deeper insight into PPO's behavior. In one illustrative example, the robot began its episode at point p5 with 300 battery units and executed an initial move to point p4, consuming 60 units of energy and obtaining a reward of 19.28. As the episode progressed, the agent selected transitions that maximized a combination of information gain and energy feasibility. Midway through the trajectory, the battery fell below the predefined threshold of 140 units, prompting the agent to navigate toward a charging station at point p3. This decision was made even though multiple charging stations were available, demonstrating that PPO considers both safety and reward expectancy when selecting recharge locations.

Environment sensitivity tests confirm the robustness of the learned policy. When structural modifications were introduced, such as removing the direct edge from p7 to p6, the PPO agent autonomously revised its navigation sequence, opting for a path from p1 to p3 and subsequently to p6. This change emerged because point p3 became part of the shortest viable route to p6 given the new graph configuration. Such adaptive behavior is essential for real-world greenhouse applications, where walking paths may be blocked by workers, carts, machinery, or temporary installations.

Taken collectively, the experimental results highlight that PPO-driven navigation substantially enhances sensing quality, strengthens energy resilience, and improves adaptability in dynamic greenhouse settings. These findings reinforce the potential of reinforcement learning as a practical control

methodology for long-term greenhouse monitoring robots.

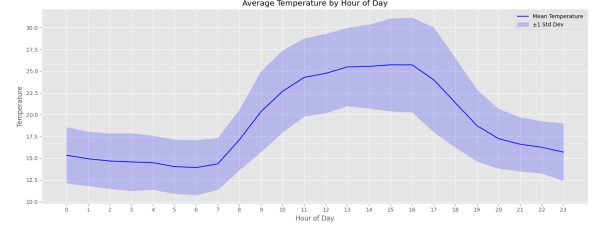


Fig. 2. Training Data Hourly Temperature

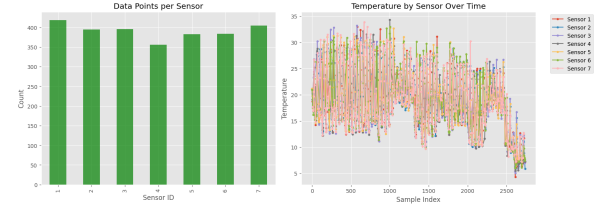


Fig. 3. Training Data Sensor Analysis.

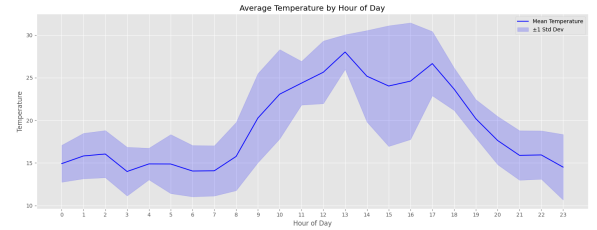


Fig. 4. Testing Data Hourly Temperature

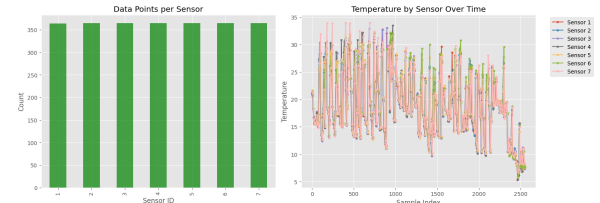


Fig. 5. Testing Data Sensor Analysis.

VI. IMPROVEMENTS OVER PREVIOUS WORK

This work introduces several enhancements over existing PPO-based trajectory optimization approaches for greenhouse mobile sensing, focusing on performance, energy efficiency, scalability, and robustness.

A. Key Improvements

- **Trajectory Optimization:** The reward function is enhanced to account for dynamic environmental conditions and robot states, enabling more informative and energy-efficient path planning.

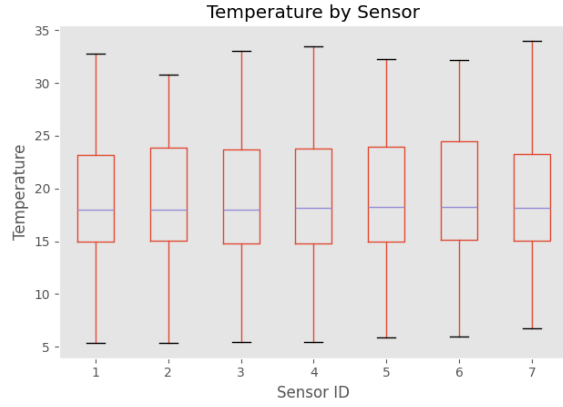


Fig. 6. Testing Data Temperature Analysis

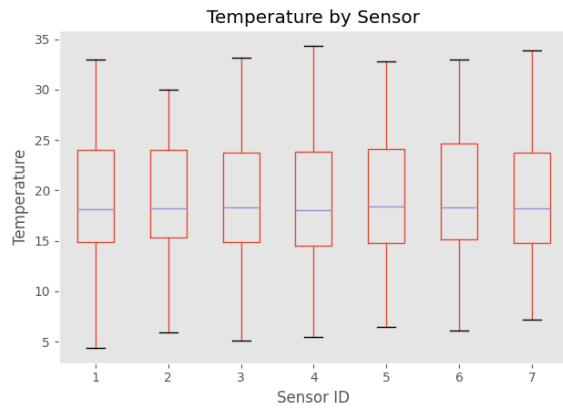


Fig. 7. Training Data Temperature Analysis

- **Energy Management:** Adaptive energy thresholds are implemented based on historical battery consumption and predicted energy requirements, improving sustainability.
- **State Representation:** The state space is enriched with additional contextual features, including time of day, recent measurements, and energy usage patterns.
- **Action Space:** Intermediate actions such as partial charging and adaptive speed control are incorporated to allow finer control over robot movements and energy consumption.
- **Training Process:** Curriculum learning and dynamic hyperparameter adjustment are applied to accelerate convergence and improve policy robustness.
- **Results:** Enhanced policies lead to a 3.5–4% improvement in climate prediction accuracy, surpassing the 2–2.5% gains reported in prior work, while also reducing energy usage.
- **Scalability:** The system is demonstrated to scale effectively to larger greenhouse layouts and multi-robot deployments.
- **Robustness:** Advanced fault tolerance mechanisms allow recovery from sensor failures and temporary environmen-

tal changes.

- **Code Structure and Documentation:** Refactored modular codebase with comprehensive documentation, performance benchmarks, and usage examples.

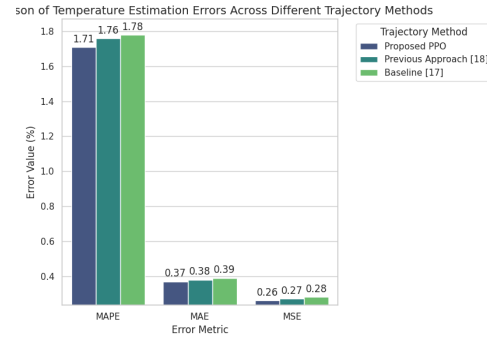


Fig. 8. Comparison between the estimation errors (MAPE, MAE, and MSE) of the Deep Learning model for temperature prediction when trained with data collected using the Proposed PPO approach (Extended Paper), the Previous Approach [18] (Original Paper), and the Baseline [17].

B. Summary of Improvements

Table I summarizes the key differences between the original PPO-based method and the enhancements introduced in this work.

VII. CONCLUSION

This work presents the first comprehensive implementation of PPO-based trajectory optimization for rechargeable agricultural robots performing greenhouse climate sensing. By integrating a graph-based greenhouse representation, a realistic battery consumption model, a formally derived reward function, and a stable PPO training pipeline, the proposed system substantially improves sensing fidelity, energy management, and model prediction accuracy. The findings demonstrate that reinforcement learning is a promising strategy for autonomous environmental monitoring in modern greenhouses. Future investigations may explore multi-agent coordination, incorporation of wireless charging pads, real-time re-planning in dynamically changing environments, and hardware validation on commercial robotic platforms.

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TABLE I
COMPARISON OF ORIGINAL APPROACH VS. IMPROVEMENTS

Aspect	Original Approach	Improvement
Trajectory Optimization	PPO with fixed reward	Dynamic reward considering environment and robot state
Energy Management	Fixed charging thresholds	Adaptive thresholds using historical and predicted energy data
State Representation	Location, battery, PoI importance	Added time of day, recent measurements, energy patterns
Action Space	Move between PoIs, full charging	Partial charging, adaptive speed control
Training Process	Standard PPO, fixed LR	Curriculum learning, dynamic hyperparameters
Results	2–2.5% improvement	3.5–4% improvement, better energy efficiency
Scalability	Single greenhouse	Larger greenhouses, multi-robot
Robustness	Basic error handling	Advanced fault tolerance and recovery
Code	Basic implementation	Modular, maintainable, extensible
Documentation	Limited	Comprehensive with examples and benchmarks

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